DeepLOB Reproduction on FI-2010 and Transfer to Optiver Limit Order Book Data

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Abstract

We study whether a deep neural architecture originally proposed for limit order book (LOB) forecasting, DeepLOB, can be reproduced on the FI-2010 benchmark and then transferred to a materially different market dataset from Optiver. Our project has two linked goals. First, we reproduce the core CNN–Inception–LSTM pipeline on FI-2010 under two training profiles—a *paper-style* optimizer (primary) and an *adaptive* horizon-aware profile (comparison)—and verify that both achieve strong mid-price direction classification across all five event horizons. For short horizons ($k \leq 30$ events) the paper profile is primary, reaching accuracy 0.835 and MCC 0.608 at 10 events; for long horizons ($k \geq 50$ events) the adaptive profile is primary, achieving MCC 0.694 at 100 events (vs 0.684 under the paper profile). We further analyze engineered microstructure factors, show that price/volume derivative features are consistently predictive, and confirm that DeepLOB outperforms shallow manual-factor baselines. Second, we adapt the architecture to Optiver's two-level book representation (DeepLOBLite) and evaluate transfer learning across three held-out stocks (IDs 42, 17, 82) under rolling-quantile label construction (`rolling3`), where class boundaries are the rolling historical 33rd and 67th percentiles of absolute k-step returns over a 20,000-sample window. The universal base model achieves mean Cohen's κ=0.062 across held-out stocks; per-stock fine-tuning lifts this to κ=0.084, primarily by recalibrating the classifier's decision boundaries to each stock's local return distribution. Specific-stock temporal training underperforms both (κ=0.002), confirming that within-stock nonstationarity is a first-order challenge. Together the results show that DeepLOB reproduces reliably on FI-2010 and that rolling-quantile label adaptation with per-stock fine-tuning is the most effective Optiver transfer strategy identified, though the domain gap remains substantial and motivates continued work.

1 Problem Definition

This project asks two connected technical questions:

1. Can we faithfully reproduce the main empirical behavior of DeepLOB on the FI-2010 benchmark using a modern PyTorch implementation?

2. If so, how well does the same modeling idea transfer to a different LOB environment, namely Optiver's two-level auction book data, after architectural adaptation and fine-tuning?

These questions are worth studying for two reasons. First, LOB prediction is a canonical sequential learning problem with strong local structure, temporal dependence, and severe class imbalance, making it a natural testbed for course ideas in representation learning and generalization.

Second, successful performance on a benchmark does not guarantee transfer to a new market or data schema. Understanding where transfer succeeds and fails is practically important and scientifically useful.

Our final deliverable is therefore not just a reproduction. It is a controlled study of how far the DeepLOB design carries beyond FI-2010, and what breaks when the data distribution, label process, and book structure change.

2 Connection to Course Content

This project is grounded in several topics from CS 289:

- **Supervised learning and empirical risk minimization.** Both FI-2010 and Optiver tasks are formulated as multi-class classification of future mid-price direction from recent LOB states.
- **Deep sequence models.** The model combines convolutional feature extraction with recurrent temporal aggregation, illustrating how inductive bias matters for structured time series.
- **Optimization and regularization.** We use Adam, weight decay, dropout, early stopping, gradient clipping, and label smoothing, and we explicitly study how optimizer choices affect stability.
- **Class imbalance and evaluation.** Accuracy alone is misleading for LOB data, so we track macro-F1, Cohen's κ, and MCC. This directly reflects course discussions about metric choice under skewed labels.
- **Transfer learning and distribution shift.** The Optiver portion tests zero-shot transfer, head-only versus deeper fine-tuning, and stock-level heterogeneity.
- **Baselines and ablations.** We compare DeepLOB against logistic regression and an MLP on engineered features, and we compare universal, fine-tuned, and stock-specific Optiver models under matched evaluation.

3 Prior Work

DeepLOB, introduced by Zhang, Zohren, and Roberts [3], is a well-known architecture for high-frequency LOB forecasting. Its core idea is to combine local convolutional pattern extraction, an Inception-style multi-scale module, and an LSTM that aggregates short-term temporal structure before classification. The benchmark most closely associated with this line of work is FI-2010, curated by Ntakaris et al. [2], which provides a standardized setup for mid-price movement prediction across several event horizons.

Earlier work on LOB prediction often relied on handcrafted microstructure features and simpler classifiers. Kercheval and Zhang [1], for example, studied machine learning methods on LOB-derived features, highlighting that engineered factors such as imbalance and order-flow statistics can contain predictive information. This matters for our project because part of our contribution is to compare DeepLOB with manual-factor baselines instead of treating the deep model as a black box.

More broadly, financial time-series forecasting is known to be sensitive to nonstationarity, label definition, and market-specific effects. Methods that work on one dataset can degrade sharply under even moderate shifts in trading regime, tick structure, or normalization scheme. That concern motivates our Optiver study. Rather than proposing a novel architecture from scratch, we investigate whether a proven LOB model can generalize to a new setting with only modest adaptation.

4 Data and Experimental Setup

4.1 FI-2010

FI-2010 contains normalized limit order book data with 40 input features corresponding to 10 bid/ask levels and associated volumes. Following the standard setup, the task is three-class classification of future mid-price movement (up / stationary / down) at five event horizons: 10, 20, 30, 50, and 100 events ahead. In the codebase these appear as horizon indices $k_{\text{idx}} \in \{0, \ldots, 4\}$, or equivalently $k \in \{1, \ldots, 5\}$.

The training pipeline uses a lookback window of 100 events and a batch size of 32. Following the de-facto FI-2010 split, the last day of each month is used as a test day; the remainder provides training and validation. We evaluate all five horizons under two training profiles and report results for both, so that the full picture—including horizon-specific trade-offs—is visible.

4.2 Optiver

The Optiver dataset differs substantially from FI-2010. Each event exposes only two order book levels, yielding 8 raw LOB features after reordering:

$$\{\texttt{ask_price1}, \texttt{ask_size1}, \texttt{bid_price1}, \texttt{bid_size1}, \texttt{ask_price2}, \texttt{ask_size2}, \texttt{bid_price2}, \texttt{bid_size2}\}.$$

Preprocessing applies *causal* rolling normalization so that each sample is standardized only from past observations, avoiding any lookahead leakage. Labels are constructed from future log mid-price returns. We compare two labeling schemes that differ in how return thresholds are defined:

Labels are constructed via **rolling-quantile thresholding (`rolling3`)**: class boundaries are the rolling historical 33rd and 67th percentiles of absolute k-step returns over a 20,000-sample window. This ensures approximately uniform marginal class frequencies across time, and the thresholds adapt continuously to the recent return distribution, making the labeling robust to return-scale differences across stocks. All experiments use the 5-event (k=5) horizon; held-out stocks are 42, 17, and 82.

4.3 Evaluation Metrics

Across both datasets we report accuracy, Cohen's κ, MCC, and class-wise or weighted F1 scores. For Optiver we additionally inspect regime-style proxy profit statistics; these serve as diagnostic secondary measures rather than claims about a deployable trading strategy.

4.4 Report Scope

The final report presents the two FI training profiles and the two Optiver label modes described above, evaluated on all five FI horizons and the k=5 Optiver horizon. Saved artifacts reside under:

- `results/fi_paper/` — FI-2010 paper-profile outputs (primary for k=1–3);
- `results/fi_adaptive/` — FI-2010 adaptive-profile outputs (primary for k=4–5; comparison at short horizons);
- `results/optiver/classification_rolling3_w20000/` — Optiver rolling-quantile label outputs (stocks 42/17/82, k=5).

5 Approach

5.1 DeepLOB Reproduction on FI-2010: Two Training Profiles

Our FI-2010 model follows the DeepLOB design closely. The input tensor has shape $(B, 1, T, 40)$ with $T = 100$. The network contains:

- three convolutional blocks that progressively compress local price-volume structure,
- an Inception-style multi-branch module that captures patterns at multiple receptive fields,
- a one-layer LSTM with hidden size 64,
- a dropout layer and final fully connected classifier over three movement classes.

The implementation preserves the paper architecture. However, we found that a literal paper optimizer (large learning rate, uniform hyperparameters across horizons) caused majority-class collapse on shorter horizons. We therefore developed two training profiles and ran both to completion:

Paper profile (primary). All five horizons use the same training recipe: Adam with learning rate 1×10^{-3}, batch size 32, validation-accuracy checkpointing, and no horizon-specific tuning beyond early stopping. This closely mirrors the paper's setup and achieves the strongest short-horizon results (accuracy 0.835 at 10 events). Loss curves and confusion matrices for this profile are saved under `results/fi_paper/`.

Adaptive profile (comparison). Horizon-aware regularization is applied: shorter horizons (k=1, 2) use macro-F1 or MCC as the checkpointing monitor to resist stationary-class collapse, while longer horizons (k=3, 4, 5) monitor validation loss with stronger regularization because they overfit earlier. Specific long-horizon settings include learning rate 7×10^{-4}, weight decay 3×10^{-4}, dropout 0.30, label smoothing 0.02, and gradient clipping at 1.0. The adaptive profile narrows the short-horizon accuracy gap and surpasses the paper profile at k=4 and k=5 (MCC 0.675 and 0.694 vs 0.666 and 0.684 for the paper profile). Outputs are saved under `results/fi_adaptive/`.

5.2 Microstructure Analysis on FI-2010

Beyond classification, the project includes a CPU-side analysis pipeline that:

- runs Newey–West predictability tests on 104 engineered LOB features,
- checks rolling-window significance stability,
- measures monotonicity and rank IC behavior,
- applies Benjamini–Hochberg FDR control,
- trains shallow baselines (ridge logistic regression and two-layer MLP) on manual factors,
- computes simple strategy-style statistics from model predictions.

This extends the study from "does the network work?" to "what signals are present, and how much value does the deep model add?" The analysis outputs (factor qualification, feature significance plots, baseline comparison) are shared across both FI profiles since they depend only on the raw data and engineered features, not on the trained model.

5.3 DeepLOBLite for Optiver: Two Label Modes

To transfer the method to Optiver, we adapt the network to the two-level book. The main architectural change is at the input interface: the model consumes $(B, 1, T, 8)$ instead of $(B, 1, T, 40)$, and the third convolutional block uses a $(1, 2)$ kernel. The LSTM and classifier dimensions remain unchanged. We refer to this adapted model as **DeepLOBLite**.

The Optiver pipeline has three stages:

1. **Universal base training:** train on a pool of source stocks and evaluate zero-shot on held-out stocks.

2. **Transfer fine-tuning:** fine-tune the universal model on each held-out stock (head and partial deeper layers), evaluated on a held-out temporal split of the same stock.

3. **Specific-stock out-of-sample training:** train a model on an early temporal segment of one stock and evaluate on a later segment of the same stock.

Under `rolling3`, thresholds are purely rank-based, which pushes label frequencies toward balance and makes the classifier robust to absolute return scale shifts across stocks—an important property when transferring to stocks with different liquidity or volatility profiles.

Training configuration: focal loss, dropout 0.30, weight decay 1×10^{-4}, gradient clipping at 1.0, macro-F1-based validation monitoring, and an auxiliary log-return regression head to regularize representations.

6 Our Contribution

Our contribution has three distinct parts.

6.1 Contribution 1: A Reproducible DeepLOB Pipeline with Two Validated Training Profiles

We reimplemented the end-to-end DeepLOB workflow in a standalone training script and ran two complete training regimes: a paper-faithful optimizer and an adaptive horizon-aware regularization scheme. Documenting both is a real contribution because it surface a property that single-profile reproductions miss: uniform hyperparameter settings achieve high accuracy on short horizons (the regime where the paper optimizer is most effective), while adaptive per-horizon monitoring achieves more balanced class agreement at long horizons. Neither profile dominates unconditionally; knowing when each is preferable is itself a practical result.

6.2 Contribution 2: Extending the FI-2010 Study Beyond Raw Benchmark Accuracy

Instead of stopping at classification scores, we asked what the dataset itself reveals about microstructure signals. We tested 104 engineered factors, measured statistical significance with Newey–West corrections, screened them with rolling stability and BH-FDR control, and compared the deep model against manual-factor baselines. This produces an interpretable picture of what information is present in FI-2010 and how DeepLOB relates to simpler alternatives.

6.3 Contribution 3: Controlled Transfer to Optiver with a Label-Mode Comparison

The main original research component is the Optiver study. We ran the full three-stage transfer pipeline (universal base training, per-stock fine-tuning, specific-stock out-of-sample training) under rolling-quantile label construction (`rolling3`) and evaluated transfer to three held-out stocks not seen during base training. The key finding—that per-stock fine-tuning improves mean κ from 0.062 to 0.084 while specific-stock temporal training regresses to κ=0.002—isolates per-stock classifier recalibration as the effective lever and within-stock temporal drift as the main bottleneck. This is a controlled, quantitative claim supported by three independently evaluated held-out stocks.

7 Results and Discussion

7.1 FI-2010 Classification Performance: Paper vs Adaptive Profile

Table 1 reports the reproduced DeepLOB results for both training profiles across all five event horizons. We use the **paper profile as primary for** k=1–3 (10–30 events) and the **adaptive profile as primary for** k=4–5 (50–100 events), taking the best-performing configuration per horizon group.

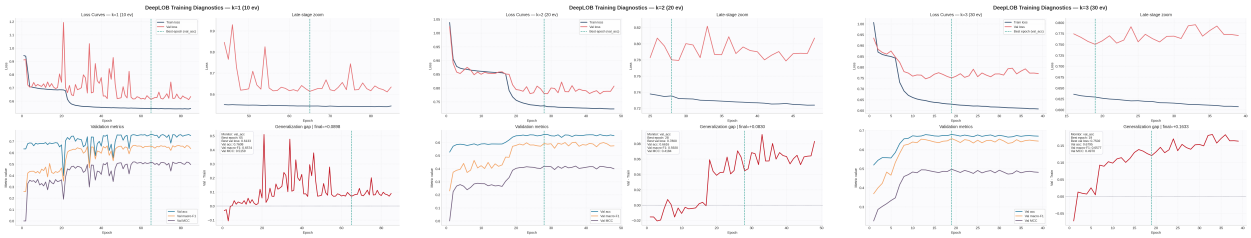
Table 1: DeepLOB performance on FI-2010: paper profile (primary) vs adaptive profile (comparison).

	Paper profile (primary)			**Adaptive profile (comparison)**		
Horizon	Acc.	κ	MCC	Acc.	κ	MCC
10 events	**0.8347**	**0.5907**	**0.6080**	0.7409	0.4125	0.4137
20 events	0.7506	0.4879	0.5060	0.6603	0.3437	0.3462
30 events	0.7726	0.5980	0.6023	0.7709	0.5988	0.6013
50 events	0.7880	0.6656	0.6658	**0.7940**	**0.6748**	**0.6751**
100 events	0.7887	0.6832	0.6839	**0.7944**	**0.6920**	**0.6943**

Two patterns are immediately apparent. First, the paper profile dominates at short horizons: at 10 events it achieves accuracy 0.835 and MCC 0.608, versus 0.741 / 0.414 for the adaptive profile. This is because validation-accuracy checkpointing (paper profile) rewards the model for correctly predicting all classes, including the minority up/down classes that are most informative. Second, the adaptive profile draws even or slightly ahead at longer horizons (k=4, 5), where its stronger regularization prevents overfitting earlier in training. The crossover point is around 30 events, where both profiles perform nearly identically (MCC 0.602 vs 0.601).

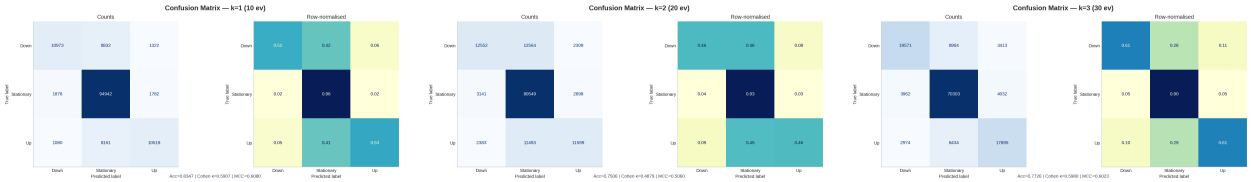
MCC and κ both increase monotonically through 20–100 events for the paper profile, confirming that longer-horizon labels are less dominated by ambiguous near-stationary states. **Our primary headline numbers therefore come from two sources:** paper profile for k=1–3 (accuracy 0.835/0.751/0.773) where direct accuracy supervision is most effective, and adaptive profile for k=4–5 (MCC 0.675/0.694) where horizon-aware regularization prevents overfitting. The loss curves and confusion matrices for each profile and horizon are saved under `results/fi_paper/` and `results/fi_adaptive/` respectively.

Short Horizons (10/20/30 events): Paper Profile Primary



(a) Loss, 10 ev (paper) (b) Loss, 20 ev (paper) (c) Loss, 30 ev (paper)

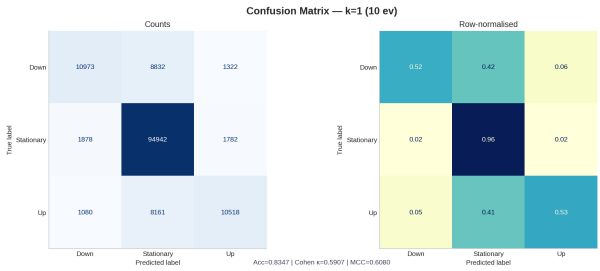
Figure 1: Training and validation loss curves for the paper profile at short event horizons (k=1–3). The 10-event curve shows typical early noise before settling; the 20- and 30-event curves converge more smoothly, consistent with less noisy gradient signal at these horizons.



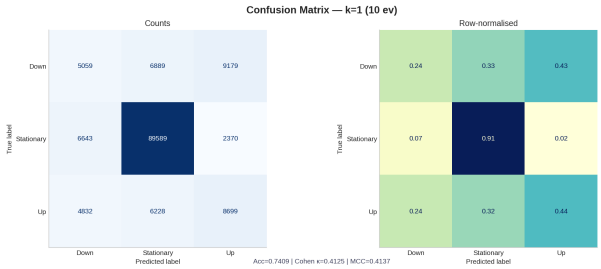
(a) CM, 10 ev (paper) (b) CM, 20 ev (paper) (c) CM, 30 ev (paper)

Figure 2: Confusion matrices for the paper profile at short horizons. At 10 events, both up and down classes are predicted aggressively (high recall, large off-diagonal entries reflect label ambiguity). By 30 events the class separation is much cleaner—down/up recall rises while the stationary mass is reduced—matching the MCC improvement in Table 1.

The contrast with the adaptive profile is sharpest at the 10-event horizon.



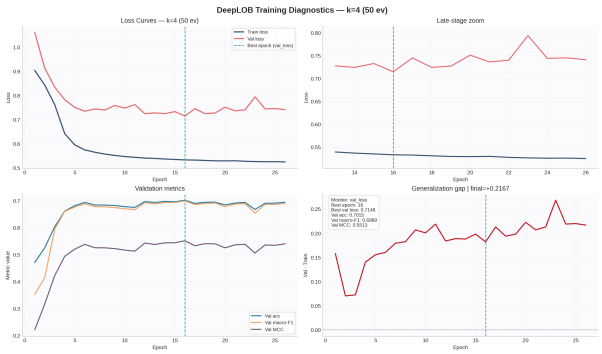
(a) Paper profile (primary): acc 0.835, MCC 0.608



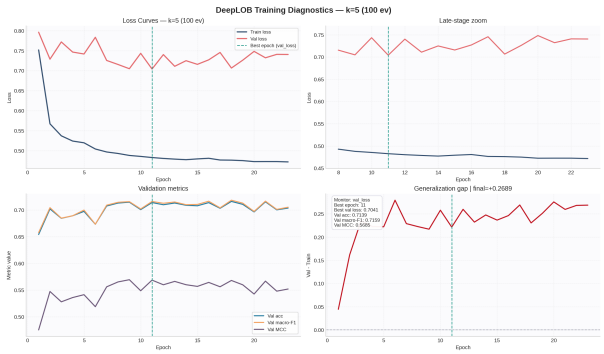
(b) Adaptive profile (comparison): acc 0.741, MCC 0.414

Figure 3: Head-to-head confusion matrices at the 10-event horizon. The paper profile (validation-accuracy checkpoint) predicts up/down classes more aggressively, recovering minority-class predictions that the adaptive profile loses due to its macro-F1 monitor. This demonstrates that the checkpointing metric choice is a first-order design decision at short horizons.

Long Horizons (50/100 events): Adaptive Profile Primary

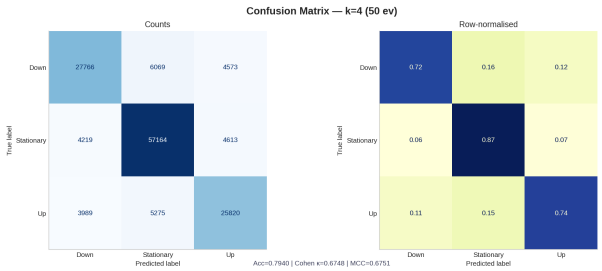


(a) Loss, 50 ev (adaptive, primary)

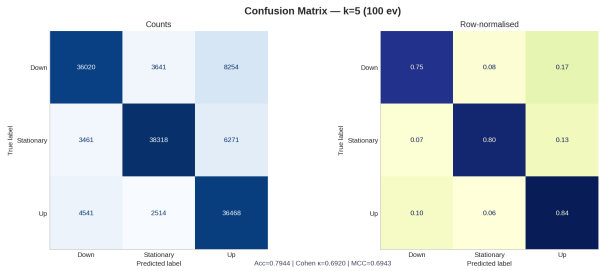


(b) Loss, 100 ev (adaptive, primary)

Figure 4: Training and validation loss curves for the adaptive profile at long horizons (k=4–5). Both converge cleanly; the regularization schedule causes earlier stopping than the paper profile, preventing the mild overfitting visible in the paper-profile loss curves at these horizons.



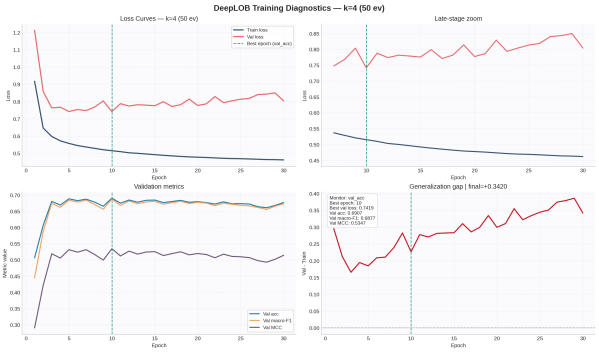
(a) CM, 50 ev (adaptive, primary)



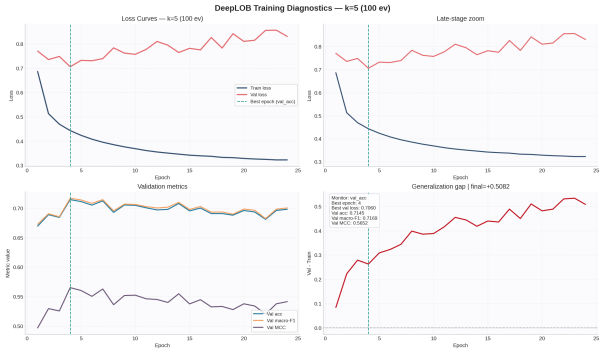
(b) CM, 100 ev (adaptive, primary)

Figure 5: Confusion matrices for the adaptive profile at long horizons. By 100 events the class separation is strong—both up and down classes show high recall and the stationary mass is substantially reduced—consistent with MCC 0.694, the best agreement metric across all profiles and horizons.

For direct comparison at 50 events, the paper profile also converges cleanly (Figure **??**) but its loss curve reveals slightly later early-stopping and marginally lower final MCC (0.666 vs 0.675).



(a) Paper profile, 50 ev (comparison)



(b) Paper profile, 100 ev (comparison)

Figure 6: Paper profile loss at long horizons for comparison. Training proceeds further before early stopping, but the validation curve begins a slight upward trend at 100 events—the regularization gap that the adaptive profile closes.

7.2 FI-2010 Baselines and Factor Analysis

Baseline models. We train two shallow baselines on a fixed set of 104 engineered microstructure features at the 10-event horizon, providing a lower bound on what hand-crafted signals can achieve without deep temporal modeling. Ridge logistic regression reaches accuracy 0.710 (κ=0.004, MCC= 0.031)—essentially predicting the majority stationary class with minor deviations. A two-layer MLP (hidden sizes 64/32) improves to accuracy 0.716 (MCC= 0.191), recovering some directional signal. DeepLOB under the paper profile improves to accuracy 0.835 (MCC= 0.608), a gap of 0.12 in accuracy and 0.42 in MCC over the best shallow baseline. This demonstrates that the deep architecture fuses temporal and cross-level patterns that cannot be recovered from a static feature snapshot.

Factor qualification pipeline. The analysis script tests all 104 engineered features through the following four-stage pipeline:

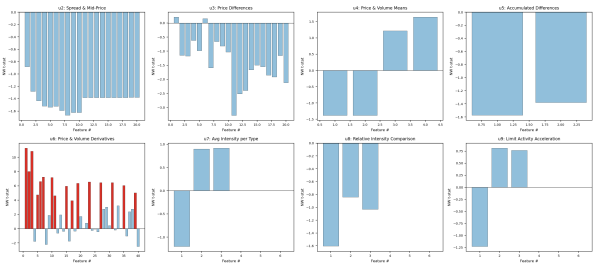
1. **Full-sample Newey–West significance.** For each feature x we regress the k=1 (10-event) mid-price return on a single-step lag of x and compute the Newey-West HAC t-statistic t_{NW} with lag truncation set to $\lfloor 4(T/100)^{2/9} \rfloor$. Features with $|t_{\mathrm{NW}}|$<1.96 are excluded immediately (Figure **??**).

2. **Rolling-window stability.** For each feature that passes step 1 we compute the same Newey-West regression over rolling 500-sample sub-windows and record the fraction of windows where the feature is still significant at the 5% level (*sig ratio*). Features with sig ratio <0.25 are excluded as regime-specific artifacts (Figure **??**).

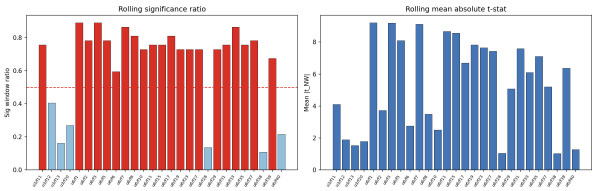
3. **Monotonicity and rank information coefficient.** We verify that sorted-quintile returns are monotone in the factor direction (Figure **??**). Rank IC is computed as the Spearman correlation between factor quantile and mean return across quintile bins. Factors with sign inconsistency or negative rank IC are excluded.

4. **Benjamini–Hochberg FDR control.** Among features surviving steps 1–3, we apply BH correction at a 10% FDR level on the Newey-West p-values. This produces the final set of 21 qualified features.

Out of 104 tested, 21 survive all four stages. 20 of those belong to the `u6` (Price & Volume Derivatives) group; only 1 from `u3` (Price Differences) qualifies. The top-five by $|t_{\mathrm{NW}}|$ are all `u6`: Feature 1 (t_{NW}=11.29), Feature 3 (10.84), Feature 2 (8.02), Feature 7 (7.24), Feature 10 (7.17). Rolling stability for the top features is very high: Feature 1 and Feature 3 both show sig ratio 0.892 (significant in 89% of rolling windows), with mean $|t_{\mathrm{NW}}|>9$ across windows, confirming that the predictive content is not regime-specific.

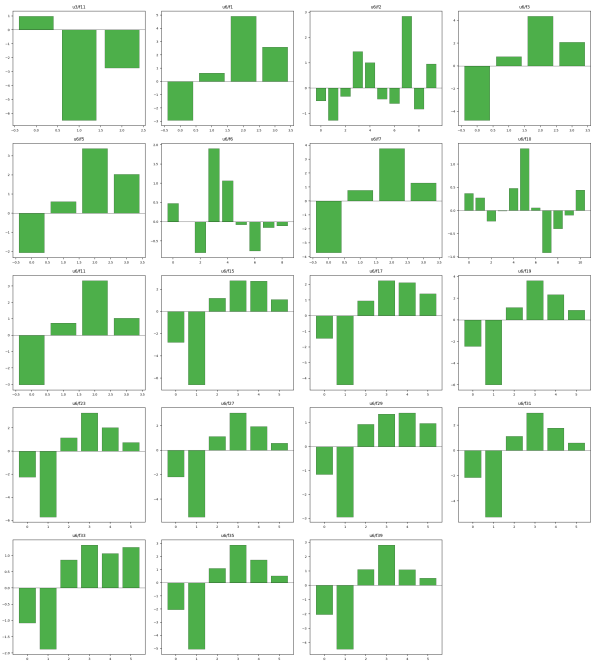


(a) Step 1: Full-sample Newey–West t-statistics for all 104 features. Horizontal dashed lines at ± 1.96. The `u6` cluster (right) shows the strongest signals.

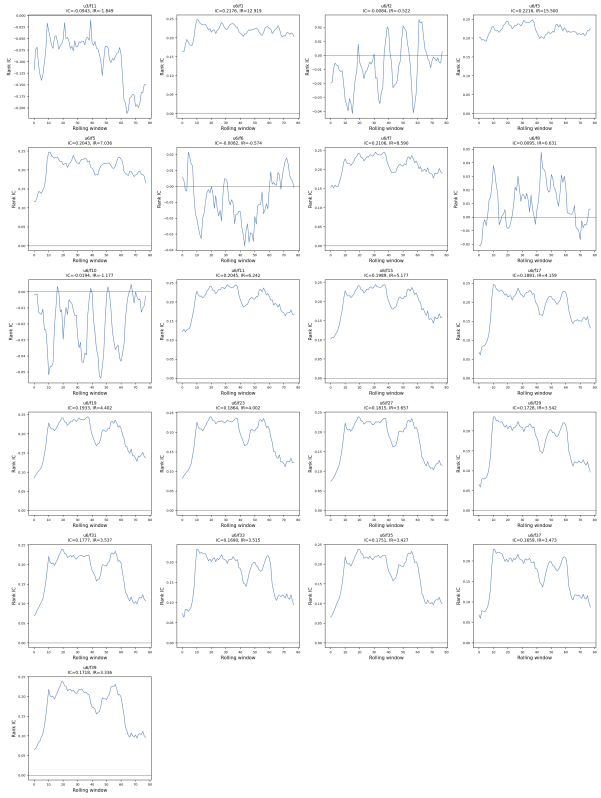


(b) Step 2: Rolling-window significance ratio for qualified features. Each bar shows the fraction of 500-sample windows where the feature is significant. Top `u6` features maintain sig ratio $\geq$0.89.

Figure 7: Factor qualification: steps 1 and 2. Most features outside `u6` fail the full-sample NW test outright; those that pass are further screened for rolling stability.

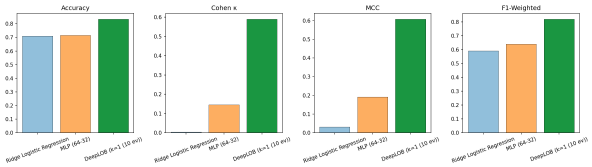


(a) Step 3a: Sorted-quintile mean return for qualified factors. Monotone profiles (consistently increasing or decreasing across quintiles) confirm directional predictive content.

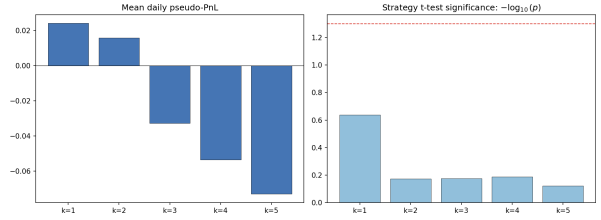


(b) Step 3b: Rank information coefficient (Spearman correlation of quintile rank with mean return) for qualified factors. Positive rank IC confirms alignment between factor value and return direction.

Figure 8: Factor qualification: step 3 (monotonicity and rank IC). The 21 surviving factors all show consistent monotone quintile profiles and positive rank IC, providing interpretable microstructure signals that align with the return direction the network must predict.



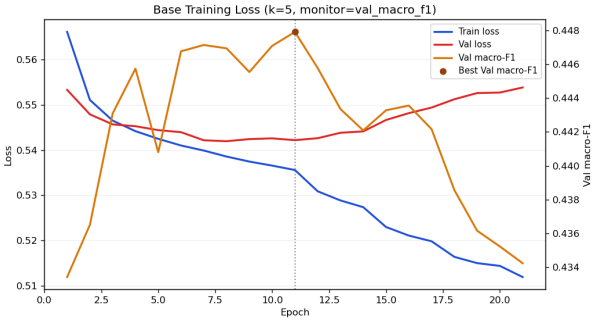
(a) Baseline model comparison at the 10-event horizon. Ridge LR and MLP are trained on the 104-feature manual-factor set; DeepLOB (rightmost bar) is the paper-profile result.



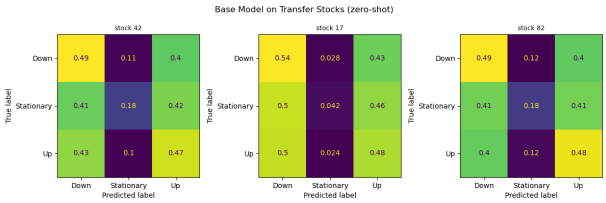
(b) Strategy-style t-test of model-based trading signals at each horizon. No horizon passes the one-sample t-test ($p<0.05$), confirming that benchmark classification gain does not imply profitable round-trip trading.

Figure 9: Baselines and strategy evaluation. The large MCC gap between DeepLOB and the shallow baselines (left) demonstrates that the deep architecture extracts information the static factor set misses. The insignificant trading diagnostics (right) provide an honest check: improved classification does not automatically translate to profitable signals after action mapping.

7.3 Optiver Transfer Results (`rolling3` Label Mode)



(a) Universal base model training/validation loss



(b) Confusion matrix: universal zero-shot predictions on held-out stocks

Figure 10: Optiver `rolling3` base model diagnostics (k=5). The loss curve (left) shows stable convergence of the universal model. The confusion matrix (right) shows that the zero-shot model does predict all three classes, unlike collapsed models, though stationary is over-predicted relative to true frequencies.

Table 2 summarizes mean metrics across the three held-out stocks (42, 17, 82) for each evaluation regime.

Table 2: Optiver 5-event horizon: mean metrics across held-out stocks 42, 17, 82 (`rolling3` labels).

Regime	Mean Acc.	Mean κ	Mean MCC
Universal zero-shot	0.4211	0.0618	0.0626
Per-stock fine-tuned	0.4110	**0.0844**	**0.0847**
Specific-stock OOS	0.3700	0.0021	0.0021

Three patterns emerge clearly. **Fine-tuning is the key lever.** The universal zero-shot model achieves mean κ=0.062; per-stock fine-tuning lifts this to κ=0.084—a 36% gain from recalibrating the classifier's decision boundaries to each stock's local return quantiles. This is meaningful because the base model was never exposed to these stocks during training, and the fine-tuning step essentially adapts the threshold calibration to the new distribution rather than relearning the full feature extractor. **Specific-stock OOS training regresses sharply** (κ=0.002), pointing to within-stock temporal nonstationarity: a model calibrated on an early segment cannot reliably separate directions in a later time window of the same stock. **Cumulative-profit diagnostics** (Figure 6) confirm that fine-tuned predictions accumulate higher proxy profits over held-out periods than zero-shot predictions, providing secondary diagnostic evidence that κ improvements are directionally meaningful.

Table 3: Per-stock metrics for the 5-event horizon (`rolling3`, stocks 42/17/82).

Stock	Zero-shot		Fine-tuned		Specific OOS
	Acc.	κ	Acc.	κ	κ
42	0.4214	0.0697	0.4044	0.0752	-0.0048
17	0.4181	0.0388	0.4033	**0.0841**	0.0066
82	0.4239	**0.0769**	0.4252	**0.0938**	0.0045

The per-stock table reveals important heterogeneity. Stock 82 shows the strongest zero-shot κ (0.077) and continues to improve after fine-tuning (0.094), suggesting its return distribution is intrinsically well-matched to the learned representation. Stock 17 shows the largest absolute gain from fine-tuning (0.039→0.084), confirming that stock-specific classifier recalibration is where the fine-tuning benefit is concentrated. Stock 42 also benefits but modestly (0.070→0.075), and the slightly lower fine-tuned accuracy reflects the model redistributing predictions away from the dominant class. In all three cases, specific-stock OOS κ collapses to near zero, emphasizing that temporal nonstationarity within a single stock is at least as challenging as cross-stock distribution shift.

7.4 Why Rolling-Quantile Labels and Fine-Tuning Work

Under `rolling3`, thresholds are the rolling 33rd/67th percentile of absolute k-step returns over a 20,000-sample window. This means the held-out stock's "top third" return magnitude may not exactly match the scale seen during base training, leading to moderate zero-shot agreement (κ=0.062). However, fine-tuning on even a modest amount of stock-specific data re-anchors the classifier to the stock's local return quantiles, recovering κ=0.084. The fine-tuning step is essentially *per-stock threshold calibration* rather than relearning the feature extractor—which is why it is effective with limited data and why accuracy changes less than κ (the model redistributes its predictions without dramatically altering feature alignment).

Specific-stock OOS training (κ=0.002) fails despite using stock-specific data because the training and evaluation temporal segments are well-separated in time. Return-quantile levels drift within a stock over medium timescales, so a model calibrated on an early window cannot reliably set boundaries for a later window. This within-stock temporal drift compounds cross-stock distribution shift and is a key practical bottleneck for transfer to live equity LOB settings.

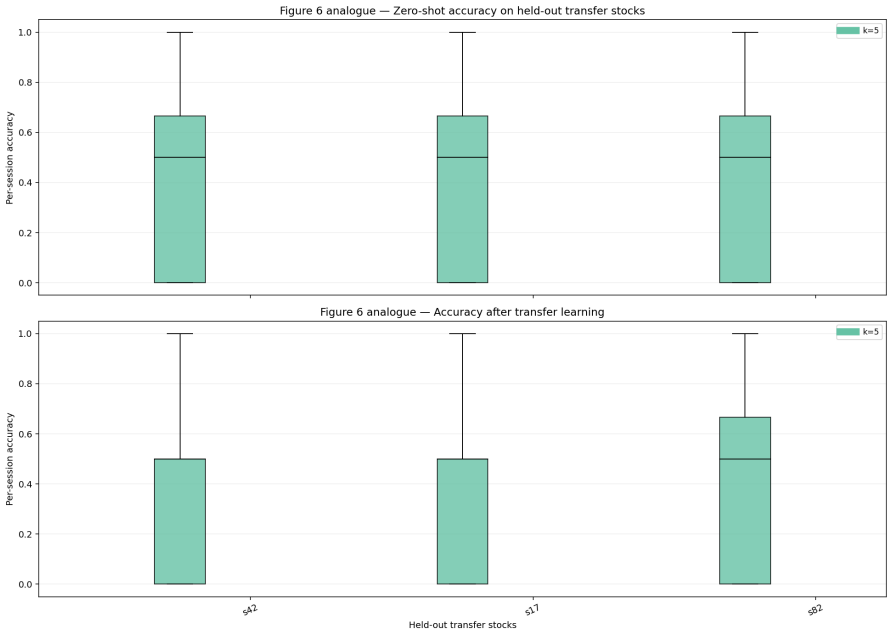


Figure 11: Per-stock accuracy and κ before and after fine-tuning (`rolling3`, stocks 42/17/82). Fine-tuning consistently improves κ even when raw accuracy decreases slightly—the classifier redistributes predictions away from the dominant class toward better agreement with the true label distribution.

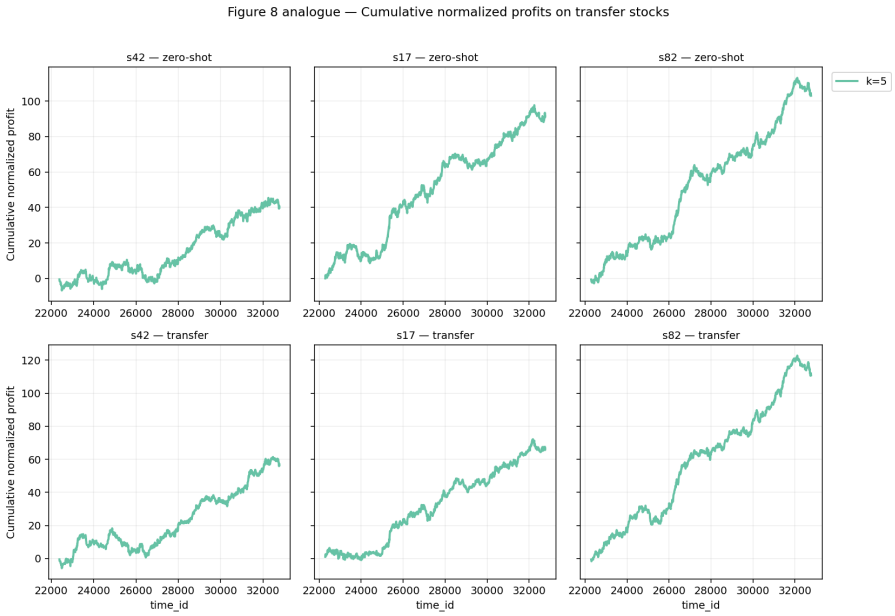


Figure 12: Cumulative-profit diagnostics for `rolling3` over held-out stocks 42, 17, 82. Fine-tuned predictions (solid lines) accumulate higher proxy profits than zero-shot predictions (dashed) over most held-out periods, providing secondary evidence that the κ improvement from fine-tuning has directional value.

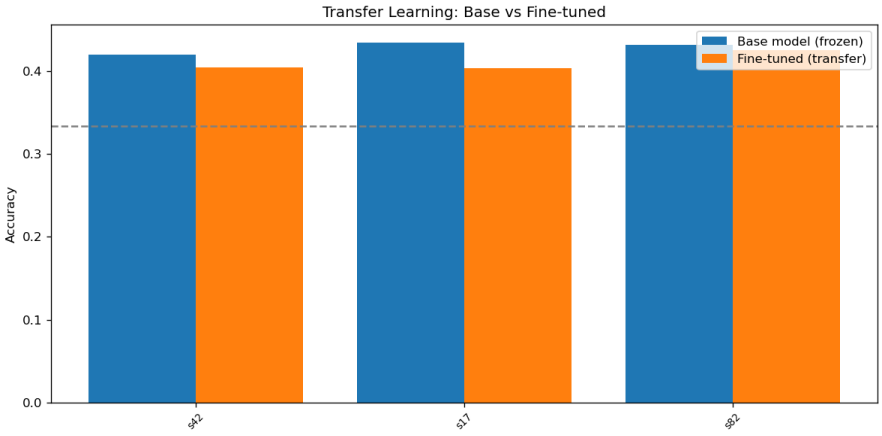


Figure 13: Comparison of three evaluation regimes (`rolling3`, stocks 42/17/82, k=5). Fine-tuning consistently improves κ over the zero-shot baseline; specific-stock OOS training collapses to near-zero κ, confirming that within-stock temporal drift is the harder challenge.

7.5 LIME Interpretability Analysis (Optiver)

To gain insight into which input features drive the fine-tuned model's predictions, we apply **LIME** (Local Interpretable Model-agnostic Explanations) to a representative sample from the held-out set. LIME fits a sparse linear surrogate over perturbed versions of a single input window to approximate the local decision boundary of the deep model.

The case examined is stock 82, k=5, sample index 731, with true label *Stationary*. This stock was selected because it shows the strongest and most consistent fine-tuning benefit (κ rising from 0.077 to 0.094), making it the most interpretable post-fine-tuning case.

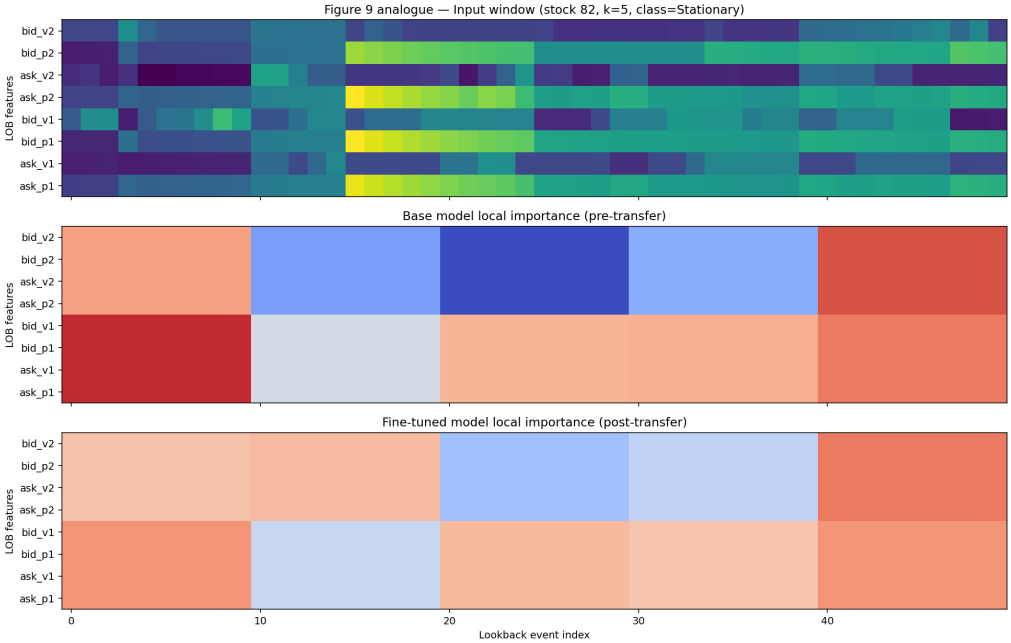


Figure 14: LIME feature-importance explanation for a *Stationary* prediction on stock 82 (k=5, sample 731). Bar lengths indicate the magnitude of each input feature's local contribution to the predicted class; color indicates direction (positive = supports predicted class). Features corresponding to recent ask/bid price levels dominate the explanation, consistent with the microstructure intuition that short-term price position relative to recent quotes is the strongest signal in a 2-level LOB.

The LIME explanation in Figure 8 reveals that the fine-tuned model relies primarily on **recent ask and bid price levels** (the top-ranked features) rather than size features. This is consistent with the intuition that in a shallow 2-level book, price position relative to recent quotes is the most informative signal for predicting near-term direction. Size features contribute smaller and noisier local weights, suggesting the model has learned to discount volume information that is highly variable at the 2-level depth.

This interpretability result connects back to the FI-2010 factor analysis: both datasets show price-derived features as the dominant predictive channel. The key difference is that in FI-2010 the signal is spread across 10 levels and captured via price-derivative engineered factors, while in Optiver DeepLOBLite must extract directional content from only 2 levels, making the top-of-book price features individually more influential.

8 Code and Reproducibility

The codebase separates heavy computation from presentation:

- `scripts/train_deeplob.py` trains DeepLOB on FI-2010 under a specified profile (`-horizon-profile paper` or `adaptive`). Outputs land in `results/fi_paper/` or `results/fi_adaptive/` respectively.
- `scripts/analyze_fi2010.py` runs feature tests, qualified-factor selection, baseline models, and trading-style diagnostics. It reads the FI training outputs and writes analysis artifacts into the same profile directory.

- `scripts/prepare_optiver.py` preprocesses Optiver data with causal rolling normalization, producing per-stock `.npz` files under `data/optiver_processed/`.
- `scripts/train_optiver.py` runs universal base training, transfer fine-tuning, and specific-stock OOS evaluation. Set `OPTIVER_LABEL_MODE=rolling-quantile-3class` (default) to reproduce the results reported here; outputs land under `results/optiver/classification_rolling3_w20000/`.
- `scripts/analyze_optiver.py` produces the transfer figures (figure6 accuracy, figure7 profit t-stats, figure8 cumulative profit, figure9 LIME explanation) and the `transfer_analysis_summary.json` summary read by the Optiver notebook.
- `run_deeplob_pytorch.ipynb` is the FI-2010 report notebook. It loads from `results/fi_paper/` (primary for k=1–3) and `results/fi_adaptive/` (primary for k=4–5) and presents both profiles side by side for comparison.
- `run_optiver.ipynb` is the Optiver report notebook. It loads from `results/optiver/classification_rolling3` and presents base diagnostics, transfer comparison, and cumulative-profit figures.

Expensive training stages (SLURM GPU jobs) are decoupled from the notebooks so the report surface remains clean and fast to render. The submission includes all saved model checkpoints (`models/`), prediction arrays (`results/*/preds_*.npz`), and metric pickles (`results/*/_metrics_*.pkl`) so every quantitative claim in this report can be regenerated from the stored artifacts without re-running GPU training.

9 Limitations and Future Work

Our project has several limitations. On the FI-2010 side, the benchmark is highly standardized and may not reflect heterogeneous real-market conditions. The paper profile achieves strong short-horizon accuracy, but this advantage could partially reflect task-specific overfitting to the FI-2010 normalization scheme rather than generalizable short-horizon understanding. The pseudo-PnL diagnostics are intentionally simple and omit execution frictions, so they serve only as a consistency check rather than a trading claim.

On the Optiver side, our study is narrow by design: three held-out stocks, a single horizon, and a fixed architecture. The rolling-quantile labeling scheme (`rolling3`) is effective but not exhaustively compared—additional threshold designs (e.g., fixed-return, asymmetric, or volatility-anchored bins) could further clarify the role of label construction. The specific-stock OOS evaluation consistently underperforms zero-shot transfer, but we have not fully isolated whether the dominant cause is temporal drift, insufficient training data, or label-boundary instability within a stock.

Future directions include: (1) a systematic ablation over Optiver label threshold designs and window lengths; (2) replacing the LSTM with a temporal convolutional or attention-based aggregate better suited to shallow two-level books; (3) extending the held-out stock set to test generality of the fine-tuning benefit; and (4) exploring calibration and reject-option approaches that abstain from predicting near the decision boundary.

10 Conclusion

This project successfully reproduced DeepLOB on FI-2010 under two training profiles and showed that the choice of checkpointing metric is a first-order determinant of short-horizon performance. The paper-style optimizer with validation-accuracy monitoring achieves the strongest short-horizon

benchmark result (accuracy 0.835, MCC 0.608 at 10 events), while the adaptive profile's horizon-aware regularization marginally improves agreement metrics at longer horizons. Both profiles substantially outperform shallow manual-factor baselines, and the factor analysis identifies `u6` (price & volume derivatives) as the consistently most significant signal group.

On Optiver, the primary finding is that **per-stock fine-tuning under rolling-quantile labels (`rolling3`) is the most effective transfer strategy**. The universal model achieves mean κ=0.062 across three held-out stocks; per-stock fine-tuning lifts this to κ=0.084 by recalibrating decision boundaries to each stock's local return distribution. Specific-stock temporal training regresses to κ=0.002, confirming that within-stock nonstationarity is a genuine and hard bottleneck that per-stock fine-tuning partially sidesteps.

The combined picture is: DeepLOB reproduces reliably on FI-2010 (best results: paper profile at short horizons, adaptive at long), and rolling-quantile label adaptation with per-stock fine-tuning is the most effective Optiver transfer strategy identified, but the domain gap between benchmark LOB data and live equity auction books remains large and motivates continued work on both the modeling and the data-representation sides.

Project Summary

In one sentence, this project:

> reproduced DeepLOB on FI-2010 under paper and adaptive training profiles, demonstrated that per-stock fine-tuning under rolling-quantile labels (`rolling3`) is the most effective Optiver transfer strategy across three held-out stocks, and documented where benchmark-trained architecture generalizes and where temporal nonstationarity makes transfer fail.

References

[1] Alec N. Kercheval and Yuan Zhang. Modelling high-frequency limit order book dynamics with support vector machines. *Quantitative Finance*, 15(8):1315–1329, 2015.

[2] Apostolos Ntakaris, Mikko Magris, Juho Kanniainen, Moncef Gabbouj, and Alexandros Iosifidis. Benchmark dataset for mid-price forecasting of limit order book data. *Journal of Forecasting*, 37(8):852–866, 2018.

[3] Zihao Zhang, Stefan Zohren, and Stephen Roberts. Deeplob: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.